



Extra Practice

1 Simplify:

$$(1) \frac{5}{4x^3} + \frac{1}{6x^2} = \underline{\hspace{2cm}}$$

$$(2) \frac{x}{x+5} - \frac{x+3}{x-2} = \underline{\hspace{2cm}}$$

2 Simplify:

$$(1) \frac{2x^2 + 1}{3x - 4} - \frac{x^2}{4 - 3x} = \underline{\hspace{2cm}}$$

A. $\frac{3x^2 + 1}{4 - 3x}$

B. $\frac{x^2 + 1}{4 - 3x}$

C. $\frac{x^2 + 1}{3x - 4}$

D. $\frac{3x^2 + 1}{3x - 4}$

$$(2) 2 - \frac{3x + 2}{2x^2} = \underline{\hspace{2cm}}$$

A. $\frac{x^2 - 2}{2x^2}$

B. $\frac{x^2 + 2}{2x^2}$

C. $\frac{4x^2 - 3x - 2}{2x^2}$

D. $\frac{4x^2 - 3x + 2}{2x^2}$

3 Simplify: $\frac{1}{x-2} - \frac{1}{y-2} = \underline{\hspace{2cm}}$.

A. $\frac{-x+y}{(x-2)(y-2)}$

B. $\frac{x-y}{(x-2)(y-2)}$

C. $\frac{-x-y}{(x-2)(y-2)}$

D. $\frac{x+y}{(x-2)(y-2)}$

4 Simplify the following expressions:

(1) $\frac{2}{2x+3} - \frac{3}{3x-2} = \underline{\hspace{2cm}}$

A. $-\frac{12x+13}{6x^2+5x-6}$

B. $\frac{13}{6x^2+5x-6}$

C. $\frac{12x+13}{6x^2+5x-6}$

D. $-\frac{13}{6x^2+5x-6}$

(2) $\frac{2x^2-1}{2x+1} - x = \underline{\hspace{2cm}}$

A. $\frac{1-x}{2x+1}$

B. $-\frac{x+1}{2x+1}$

C. $\frac{x+1}{2x+1}$

D. $\frac{x-1}{2x+1}$

5 Simplify: $\frac{4}{9x^2 - 4} - \frac{1}{3x - 2} = \underline{\hspace{2cm}}$.

A. $-\frac{1}{3x - 2}$

B. $\frac{1}{3x + 2}$

C. $\frac{1}{3x - 2}$

D. $-\frac{1}{3x + 2}$

6 Simplify: $\frac{2x}{x^2 - y^2} - \frac{1}{x - y} = \underline{\hspace{2cm}}$.

A. $x + y$

B. $\frac{1}{x - y}$

C. $x - y$

D. $\frac{1}{x + y}$

7 If $a + \frac{1}{b} = 1$, $b + \frac{1}{c} = 1$, then $c + \frac{1}{a} = \underline{\hspace{2cm}}$.

A. 1

B. $\frac{1}{b - 1}$

C. $\frac{1}{a(b - 1)}$

D. $\frac{1}{a - 1}$

8 Simplify: $\frac{4}{1-2m} - \frac{3+2m}{1-4m^2} \div \frac{1}{2m+1} = \underline{\hspace{2cm}}.$

9 Simplify: $\frac{y}{x^2-y^2} \div \left(1 - \frac{x}{x+y}\right) = \underline{\hspace{2cm}}.$

10 Simplify $\left(a + \frac{1+a}{2a-1}\right) \div (2a^2+1)$, and then find the value when $a = \sqrt{2} + \frac{1}{2}$.

The value is $\underline{\hspace{2cm}}.$

4 Solve the following rational equations:

(1) $\frac{x+1}{3} + \frac{x+2}{2x} = 1$

$x = \underline{\hspace{2cm}}$

(2) $\frac{x-1}{x+3} - \frac{1}{x+2} = 1$

$x = \underline{\hspace{2cm}}$

5 Solve the rational equation $\frac{3}{x+2} = 1 - \frac{x^2+3x+2}{8}$.

6 Solve the following rational equation: $\frac{1}{1-x} - 1 = \frac{x-1}{2}$

$x = \underline{\hspace{2cm}}$

7 Solve the following rational equation: $\frac{4}{x^2 - 16} + \frac{x}{x + 4} = 1$

$x = \underline{\hspace{2cm}}$.

8 Solve the rational equation $\frac{x - 1}{x + 1} = 1 - \frac{4}{1 - x^2}$.

A. $x = 1$

B. $x = -1$

C. $x = \pm 1$

D. No solution

- 9 Of the 400 balls in a large bag, 85% are red and the rest are blue. How many of the red balls must be removed so that 75% of the remaining balls are red?

A. 85

B. 100

C. 120

D. 150

E. 160

6 Solve the following rational equation: $\frac{4x^2 - 9}{2} - \frac{2x - 3}{2x} = -1$

$x = \underline{\hspace{2cm}}$

7 Solve the following rational equation: $\frac{x^2}{x^2 + 4x + 4} - \frac{1}{x + 2} = 1$

$x = \underline{\hspace{2cm}}$

8 Solve the following rational equations:

(1) $\frac{x - 1}{2} - \frac{3 - x}{1} = \frac{x^2 - 4x + 3}{1}$

(2) $\frac{2x - 4}{3} - \frac{x^2 + x - 6}{2x} = \frac{x + 3}{3}$

- 9 Of the 600 balls in a large bag, 80% are red and the rest are blue. How many of the red balls must be removed so that 75% of the remaining balls are red?

A. 50 B. 75 C. 100 D. 120 E. 150

- 10 To protect arable land, a certain area needs to convert 1500 acres of forest back to arable land. It is known that the number of acres Team A can convert back to arable land per day is 1.2 times that of Team B. If each team completes the task alone, Team A would take 5 days less than Team B. How many acres can Teams A and B complete per day?

8 Solve the following quadratic equations:

(1) $2x^2 - 3x - 2 = 0$

$x = \underline{\hspace{2cm}}$ and $x = \underline{\hspace{2cm}}$.

(2) $3x^2 - 5x + 2 = 0$

$x = \underline{\hspace{2cm}}$ and $x = \underline{\hspace{2cm}}$

9 Let d and e denote the solutions of $4x^2 + 4x - 3 = 0$. The value of $(d + \frac{1}{2})(e + \frac{1}{2})$ is $\underline{\hspace{2cm}}$.

10 Let f be a function for which $f(z) = 81z^2 + 9z + 1$. The sum of all values of z for which $f(z) = 7$ is $\underline{\hspace{2cm}}$. (Adapted from 2000 AMC 12 Problems, Question #15)

Lesson 16 Practice

Basic Practice

1 Solve the following quadratic equations by completing the square:

(1) $x^2 - 8x + 10 = 0$

$x = \underline{\hspace{2cm}}$ and $x = \underline{\hspace{2cm}}$.

(2) $x^2 - 2x - 3 = 0$

$x = \underline{\hspace{2cm}}$ and $x = \underline{\hspace{2cm}}$.

2 Solve the following quadratic equations by completing the square:

(1) $2x^2 + 5x + 1 = 0$, $x = \underline{\hspace{2cm}}$ and $x = \underline{\hspace{2cm}}$.

(2) $4x^2 = 6x + 3$, $x = \underline{\hspace{2cm}}$ and $x = \underline{\hspace{2cm}}$.

3 Solve the following quadratic equations by the quadratic formula:

(1) $5x^2 - 5x + 1 = 0$, $x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$.

(2) $x^2 + 2x - 7 = 0$, $x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$.

(3) $2x^2 - 5x - 3 = 0$, $x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$

4 Solve the following quadratic equations by the quadratic formula:

(1) $2x^2 - 2x - 5 = 0$

$x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$.

(2) $3x^2 - 4x - 2 = 0$

$x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$.

- 5 Apply the Quadratic Formula to solve the following problem and verify your answer by Completing the Square.

There is a positive integer n such that $(n+3)! + 3(n+2)! = (n+1)! \cdot 21$. The value of n is _____. (Adapted from 2019 AMC 10B Problems, Question #6)

- 6 Find the value of the discriminant and use the value to determine whether the quadratic equation has two different roots, a double root or no real root.

(1) $x^2 - 5x + 3 = 0$

(2) $x^2 - 4x + 6 = 0$

- 7 If the quadratic equation $(k+1)x^2 + 3x + 1 = 0$ in terms of x has two different solutions, the range of k is _____.

- 8 Find the value of the discriminant and use the value to determine whether the quadratic equation has two different roots, a double root or no real root.

(1) $2x^2 + 5x - 6 = 0$

- A. Two different roots B. A double root C. No real root

(2) $3x^2 + 7x = -7$

- A. Two different roots B. A double root C. No real root

(3) $3x^2 - 5x = -2$

- A. Two different roots B. A double root C. No real root

- 9 If the quadratic equation $x^2 + 2x - 2m = 0$ in terms of x has two different solutions, the range of m is _____.

2 Solve the quadratic equation by the quadratic formula: $x^2 + 2x - 1 = 0$.

3 Solve the following quadratic equations by completing the square:

(1) $2x^2 - 6x + 1 = 0$

(2) $2x^2 - 3x - 1 = 0$

4 Solve the following quadratic equations by completing the square:

(1) $x^2 - 8x - 4 = 0$, $x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$.

(2) $x^2 - 6x - 5 = 0$, $x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$.

- 5 Solve the following quadratic equations by the quadratic formula:

(1) $x^2 - 6x - 2 = 0$, $x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$.

(2) $x^2 + 2x - 3 = 0$, $x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$.

(3) $x^2 + 5x - 3 = 0$, $x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$.

- 6 Find the values of the discriminants and use the values to determine whether the quadratic equations have two different roots, a double root or no real root.

(1) $3x^2 + 2x - 1 = 0$

(2) $\frac{1}{2}x^2 - 7x + \frac{49}{2} = 0$

(3) $-3x^2 + 6x - 5 = 0$

7 Use proper methods to solve the following quadratic equations:

(1) $4x^2 - 8x + 1 = 0$, $x = \underline{\hspace{1cm}}$, and $x = \underline{\hspace{1cm}}$.

(2) $3x^2 + 3x - 10 = 0$, $x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$.

8 If the quadratic equation $(k - 3)x^2 + 3x - 2 = 0$ in terms of x has two different solutions, the range of k is $\underline{\hspace{1cm}}$.

A. $k > \frac{15}{8}$

B. $k \geq \frac{15}{8}$

C. $k > \frac{15}{8}$ and $k \neq 3$

D. $k \geq \frac{15}{8}$ and $k \neq 3$

- 9 If the quadratic equation $x^2 + 5x + 2c = 0$ in terms of x has a double root, the value of c is _____.
- 10 The sum of the reciprocals of the roots of the equation $2020x^2 + 2019x + 2018 = 0$ is _____. (Adapted from 2003 AMC 10A Problems, Question #18)

- 3 There is an advancing 8-person queue where each person's height is different and they are arranged in a ascending order from the shortest to the tallest. Now they need to make a 2-by-4 formation, with 2 rows and 4 columns. If people in the same row from left to right are still arranged in a ascending order from the shortest to the tallest, and it is also required that for each pair of adjacent people in the same column, the person in front should be shorter than the one behind. Then, there are _____ different ways to arrange the 8 people.
- 4 Consider integers 0, 1, 2, 3, 4, 5:
- (1) We can form _____ three-digit numbers without repeating digits.
- (2) We can form _____ three-digit numbers with repeating digits allowed.
- (3) We can form _____ four-digit numbers that are greater than 3000 but less than 5421 without repeating digits.

- 9 Four people, namely A , B , C and D , are passing the ball. The rule is that no one can pass the ball to themselves. The ball starts in the hand of A and after 5 passes, it ends up in the hand of B . The total number of different passing methods is _____.



Extra

- 1 Three-digit numbers satisfying $a + c = 2b$ amount to () .
- A. 16 B. 36 C. 45 D. 49

- 2 Five friends a , b , c , d , and e gathered in the park and greeted each other with handshakes. It is known that a shook hands 4 times, b shook hands 1 times, c shook hands 3 times, and d shook hands 2 times. So far, e shook hands _____ times.

- 6 All students in class either join baseball club or basketball club, or join both. Among them, 16 students join basketball club, 23 students join baseball club, and 5 students join both. This class has _____ students in total.
- A. 23 B. 30 C. 34 D. 16

- 7 In the diagram, the area of each circle is 1 cm^2 . The area common to two overlapping circles is 0.1 cm^2 . What is the area of the region covered by the five circles shown? (Adapted from 2014 Math Kangaroo Problems, Level 7-8, Question #12)



- A. 4 cm^2 B. 4.6 cm^2 C. 4.2 cm^2 D. 4.4 cm^2 E. 4.5 cm^2
- 8 In a class of 50 pupils, 32 pupils like to eat fish, 40 pupils like to eat chicken and 46 pupils like to eat vegetables. What is the least number of pupils who like all three types of food?

- 4 Compute the values of the following expressions.

(1) $2({}_6P_4) - ({}_3P_2) = \underline{\hspace{2cm}}$.

(2) ${}_{12}C_2 = \underline{\hspace{2cm}}$.

(3) $({}_7P_1) - ({}_7C_1) = \underline{\hspace{2cm}}$.

- 5 There are 10 people in a party and each shakes hand with everyone else only once. How many handshakes take place during the party?

- 6 How many different six-digit numbers can be formed using all of the following digits: 4, 4, 6, 6, 3, 3?
- 7 There are six people, A, B, C, D, E , and F , who are divided into three groups: red, blue, and green. Each group consists of two people to play a game. There are a total of _____ different ways to divide them.
- 8 During the 40 Weifang International Kite Festival, a school sends 5 people to participate in 6 days of volunteer service activities. Among them, Jack continuously participates for 2 days, while others participate for 1 days each. The number of different arrangement methods is _____.



- 9 DeepSeek software is a deeply exploratory artificial intelligence software developed in China. Now rearrange the letters of the word DeepSeek. The number of different arrangements where the letters e are not adjacent is _____.

A. 24 B. 120 C. 576 D. 2880

- 10 A, B, C, D, E, F form a line, satisfying the conditions that A, B are adjacent, C, D are not adjacent, and E does not stand at the two ends. How many different arrangements are there?

A. 48 B. 96 C. 72 D. 144 E. 288



Extra Practice

- 1 How many ways can 8 different books be arranged on a shelf that can hold 3 books?

- 7 The stationery store has 8 different colors of markers. Peter is allowed to buy exactly 2 markers of different colors. How many different color combinations does he have?
- A. 8 B. 16 C. 28 D. 36 E. 56
- 8 5 boys and 4 girls are going to attend a stage show. There are exactly 5 different male characters and 4 different female characters in the show. There are _____ ways to assign the roles to them.

- 9 A team has 5 boys, 4 girls. Now we need to pick 3 people from this team to do the cleaning.

(1) If we are to pick at least 2 boys, there are _____ ways of picking.

(2) If we are to pick at least 1 girls, there are _____ ways of picking.

- 10 There are 4 students and 2 teachers standing in a row for a photo. The number of different arrangements where the 2 teachers must stand together is _____.